

Research on Improvement of Configurable I2C controller IP Core

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Abstract—For ease of the building of our own SoC system and realization of more convenient I2C communication between the system and external devices, this paper divided the functions in accordance with I2C bus data transmission principle, and then designed various functional modules based on the Verilog HDL language, which improved the configurability and flexibility of the I2C controller IP core. To advance the use of application software developers efficiently, the interaction signals of information inside and outside the IP core were simplified, and the correctness of its data transmission is verified by means of hardware platform.

Keywords—I2C controller, function division, Configurability, simplification, IP core

I. INTRODUCTION

I2C bus is a bus protocol with simple structure, few wires, and wide application. There are many IP cores in I2C bus, but all of them have independent intellectual property, and most of them have to pay if used directly, and the IP cores are designed differently for different applications [1]. With the improvement of product functions and the development of digital design technology, there is a need to design our own IP cores in order to build our own SoC systems [2]. Wang Ronghua [3] designed a configurable IIC protocol controller IP core based on the Avalon bus interface, there are many signal interactions among modules in the design; Ye Kuntao et al. [4], Zhang Suping et al. [5], and Zhu Chengcheng et al [6] have studied the portability and configurability of the IIC bus IP core, improved the portability of the product, but Ye Kuntao et al., Zhang Suping et al., however, there are many signals of external communication in the IIC bus controller with configurable ports designed by Ye Kuntao et al. and Zhang Suping et al., which is bad for developers to develop application software; the IIC bus IP core designed by Zhu Chengcheng et al. can only communicate with fixed peripheral devices.

This paper designed the I2C controller IP core based on Verilog HDL language, divided its functional modules in order to enhance its configurability, and simplified the signals of information interaction inside and outside the IP core, and finally verified the correctness of its data transmission through the hardware platform.

II. I2C BUS

A. Features of I2C Bus

The I2C bus is a simple two-wire interface introduced

by Philips Company, the full duplex data transmission is completed by the serial clock line SCL and the serial data line SDA. Because it is characterized by few wires, allowing multi host control, having bus arbitration and synchronization, etc. [7], which is widely used in EEPROM, LCD drivers, RAM and I/O ports, and has become a worldwide industry standard.

The I2C bus transmits information among devices through two bidirectional lines (clock line SCL and data line SDA.) The output stages of SDA and SCL must be open-drain or open-set electrode outputs, it is connected to the positive power supply by external pull-up resistor to fulfil the "wired and" function. When the bus is idle, SDA and SCL are high level. Each device on the bus has a unique address, both as a sender or receiver of data. Master is used to control the bus to realize data transmission, generate the clock pulse required for transmission, and the addressed device can only be in slave mode.

The data transmission rate on the I2C bus can be up to 100 kbit/s in standard mode, up to 400 kbit/s in fast mode and up to 3.4 Mbit/s in high-speed mode.

B. Data Transmission of I2C bus

The data transmission unit on the I2C bus is 8 bits, after sending 8 bits of data, the receiver will return a 1-bit status value (ACK/NoACK), so the I2C bus is a transmission unit with a total of 9 clock cycles, and each data bit transmitted generates a clock pulse.

The data transmission process is: first the host generates the start bit, and then transmits the first byte, the first bit transmitted in the 8-bit data is the highest bit of the data: MSB, and the lowest bit, LSB, is the read-write indicating bit. 1 represents read by the host, 0 represents write, the high 7-bit address can make the host to address 128 slave devices, each chip defines the address differently, and the specific chip manual should be consulted when programming.

When the slave receives the first byte, it sends a response bit, after the host receives the response bit, it sends the second byte of data. After the data is sent, the end bit is generated and the data transmission is ended. When transmitting the data, the SDA is allowed to switch only when the clock SCL is low level, and the SDA must be stable when SCL is high level, at this time the level of the SDA is the value of the bus transfer, as shown in Figure 1.

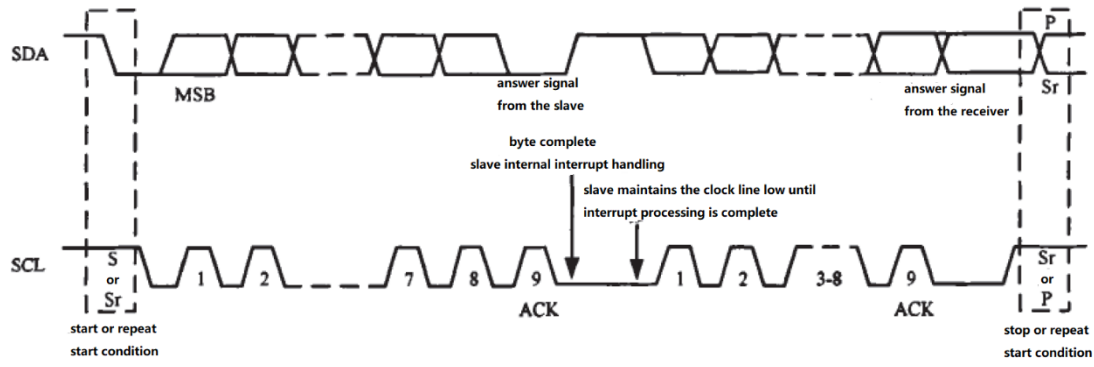


Fig.1 Data transmission on the I2C bus

III. IMPLEMENTATION OF I2C CONTROLLER IP CORE

According to the transmission principle of I2C bus data, this paper designed a configurable I2C controller IP core based on the verilog HDL language, which consists mainly of four.v files: I2C_user.v, clk_div.v, i2c_interface_v1.v and i2c_RW_Controller.v.

A. Division of Functional Module

In order to enhance the configurability of the I2C controller IP core, this paper divides the I2C controller IP

core into three functional sub-modules: the clock module clk_div, the underlying driver module i2c_interface_v1 and the read-write control module i2c_RW_Controller, as shown in Fig.2. The clock module is used to generate the operating clock of the I2C bus; the underlying driver module uses a state machine to implement the data transmission timing of the I2C bus; the read-write control module is used to generate control words and data of various operations of I2C bus, among which the underlying driver module is applicable to any type of I2C chip.

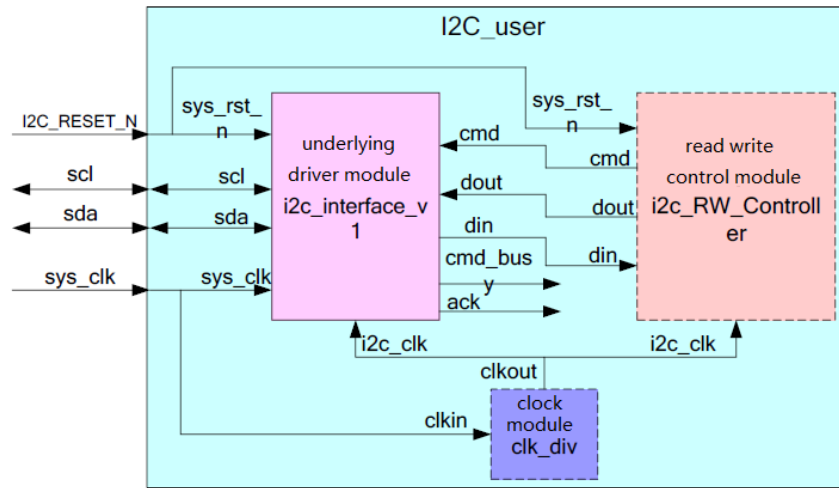


Fig.2 Interconnection relationships among the modules

The top-level file of these three submodules is I2C_user.v, which contains four interfaces: clock, reset, serial clock scl and serial data sda.

```
module I2C_user(
  input sys_clk,
  input I2C_RESET_N,
  inout scl,
  inout sda
```

```
)
(
  input clkin ,
  output reg clkout = 0
);
clkout rate = clkin rate / (DIV-2), in order to make the
designed I2C controller IP core more flexible, this paper
refined the frequency division variable DIV, the user only
needs to modify the value of DIV to change the clkout
frequency.
```

B. Clock Module clk_div

The clock module clk_div contains 2 interfaces and 2 constants to generate a clock signal at 4 times the SCL clock frequency.

```
module clk_div #
(
  parameter DIV = 22,
  parameter LEN = 5
```

C. Underlying Driver Module i2c_interface_v1

The underlying driver module i2c_interface_v1 contains 10 interfaces, which implement the data transmission timing of the I2C bus by using the state machine, this module is suitable for any type of I2C chip.

1) Interfaces

```

module i2c_interface_v1
(
    input          i2c_clk,
    input          sys_clk,
    input [3:0]    cmd,
    output reg     cmd_busy=0,
    output reg     ack,
    input [8:0]    dout,
    output reg [8:0] din,
    inout wire     scl,
    inout wire     sda,
    input          sys_rst_n
);

```

Description of the function of each interface.

- a) i2c_clk: reference clock, from the clock module clk_div.
- b) sys_clk: system clock, backup.
- c) cmd: status indication, from the read-write control module i2c_RW_Controller.

4'b1000: start
4'b0100: write operation

- 4'b0010: read operation
- 4'b0001: end
- 4'b0000: idle
- d) cmd_busy: 1 indicates that the i2c is carrying out operation; 0 indicates that the i2c is idle.
- e) ack: the 1-bit status value returned by the receiver.
- f) dout: data of the write operation, including ACK, from the read-write control module i2c_RW_Controller.
- g) din: data of the read operation, including ACK.
- h) scl: serial clock line of the i2c bus, scl= i2c_clk/4, each chip allows different maximum SKL, see chip manual for details.
- i) sda: serial data line of the i2c bus.
- j) sys_rst_n: control i2c bus reset.

2) Flowchart of The Underlying Driver Implementation

The underlying driver module uses a state machine to implement the timing of the start, end, read operation and write operation of the I2C bus, the flowchart of which is shown in Fig.3.

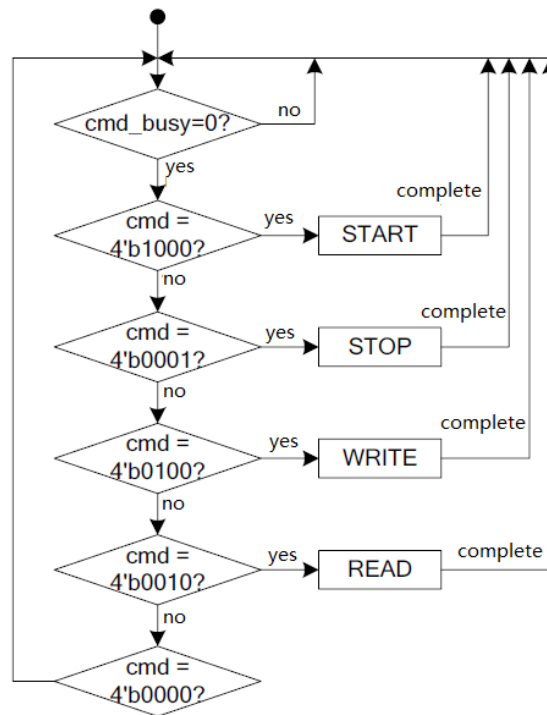


Fig.3 Flow chart of the underlying driver module

D. Read-Write Control Module i2c_RW_Controller

In order to enhance the configurability of the designed I2C controller IP core and make it more flexible, this paper separately refines the control word cmd, write data dout and read data din used to generate various operations of I2C bus, the user only needs to change the value of dout according to the specific slave device manual, without affecting the other sub-modules. This read-write control module i2c_RW_Controller contains 5 interfaces.

1) Interface

```

module i2c_RW_Controller(

```

```

    input i2c_clk, //input, clock
    output reg [3:0] cmd, //output, signs of start, end,
    read, write operations
    output reg [8:0] dout, //output, data of write
    operation
    input [8:0] din, //input, data of read operation
    input sys_rst_n //input, reset
);

```

2) Flow Chart of the Read-Write Control Module

This module contains a 20-bit counter, through which control the times of start, write, read and end. The flow

chart of byte write operations is shown in Fig.4.

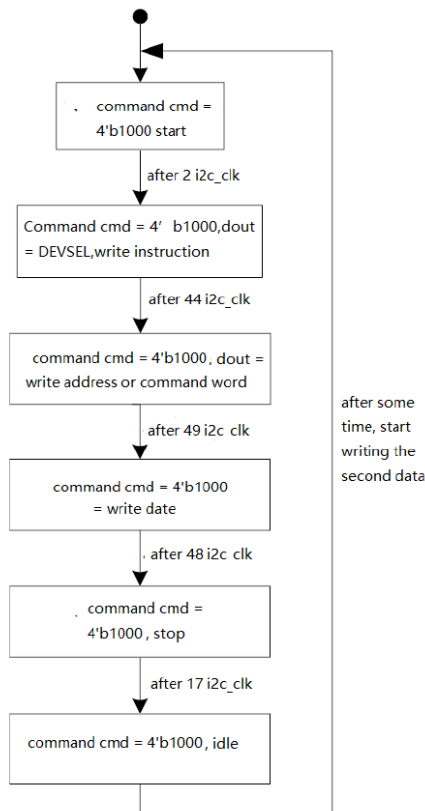


Fig.4 Operation flow of byte write

The flow chart of the random read operation is shown in Fig.5.

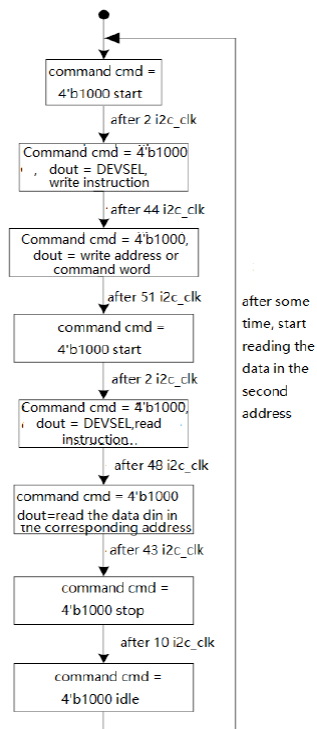


Fig.5 Random read operation flow

IV. HARDWARE VERIFICATION OF I2C CONTROLLER IP CORE

In order to verify the correctness of the design in this paper, a homemade hardware board is used for verification, FPGA adopts the XC6SLX150T of xilinx Company, the CDCL6010 is a 1-drive-10 clock driver chip, and the FPGA sets the phase locked loop on the CDCL6010 through the I2C interface, so as to meet the needs of different frequencies on the board, the verification platform is shown in Fig.6, among which FPGA is the master device, the CDCL6010 is the 2 slave devices, and the access of the 2 CDCL6010 pieces is distinguished by the address lines ADD0 and ADD1.

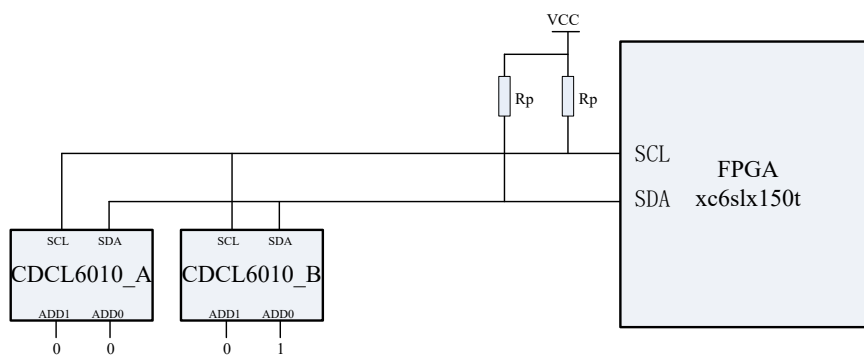


Fig.6 Hardware verification platform

The chipScope in ISE is used for simulation, the simulation results of two slave devices' write operations are

shown in Fig.7 and Fig.8, and the simulation results of slave devices' read operations are shown in Fig.9.

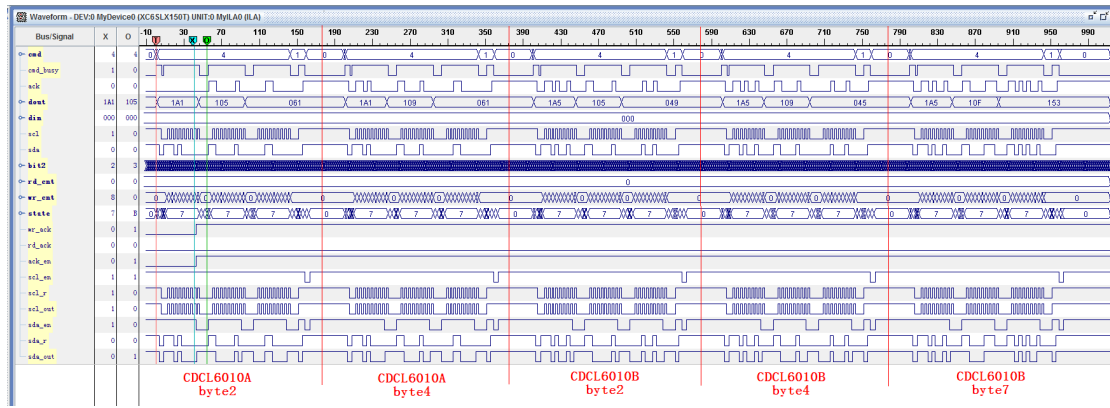


Fig.7 Simulation diagram of write operation to two slave devices (1)

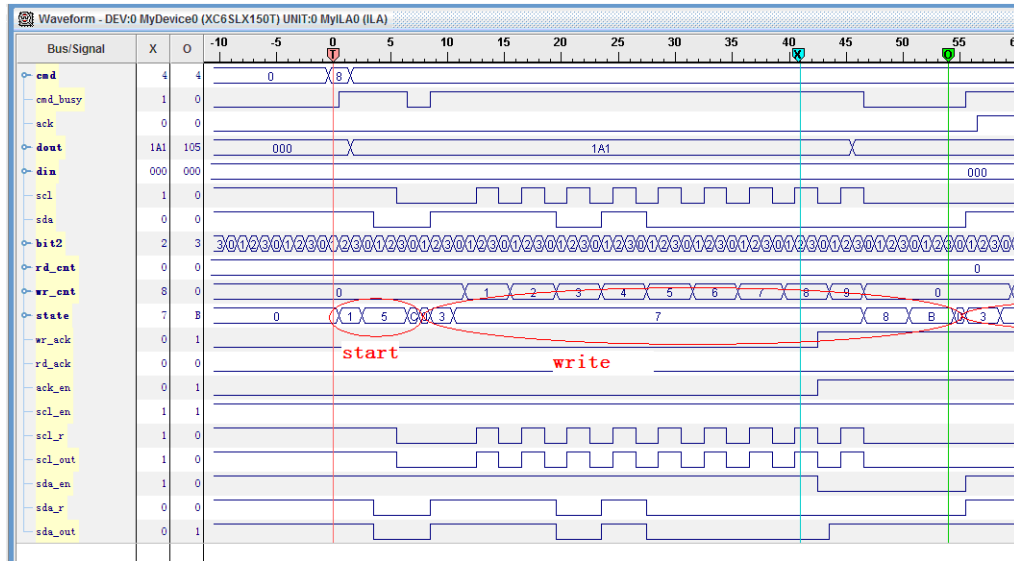


Fig.8 Simulation diagram of write operation to two slave devices (2)

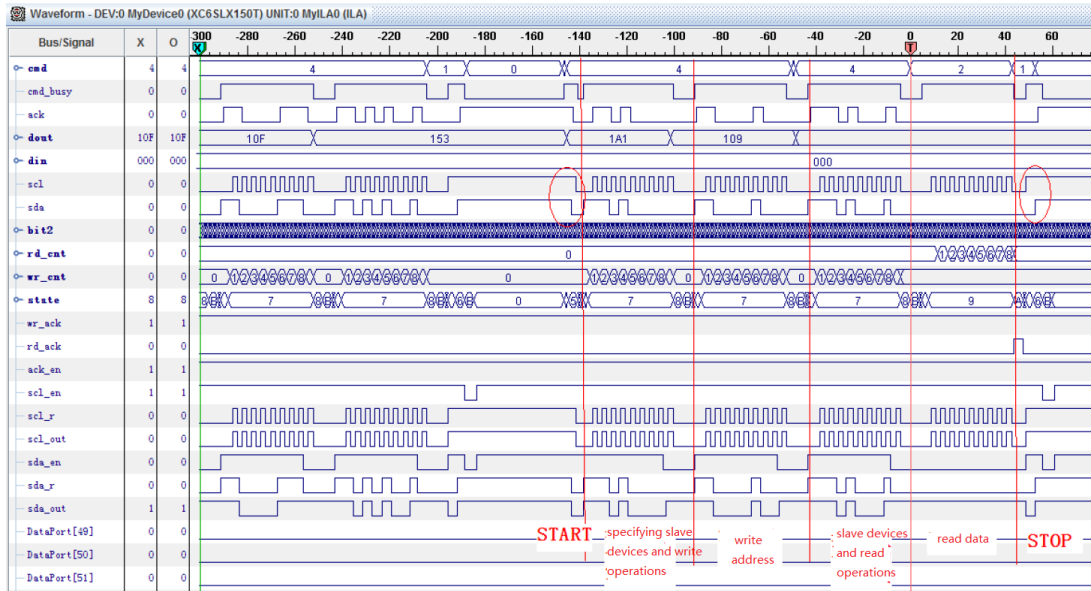


Fig.9 Simulation diagram of read operation on a slave device

According to comparison of Fig.7, Fig.8, Fig.9 with Fig.1, it shows that the I2C controller IP core designed in this paper conforms to the data transmission timing of the I2C bus and proves the correctness of the design.

V. CONCLUSION

This paper designed the I2C controller IP core based on the Verilog HDL language, divided it into three functional modules in order to enhance its configurability, simplified

the information interaction signals inside and outside the IP core for the ease of application software developers, finally, carried out the simulation verification on data transmission timing through the hardware platform, which proves the correctness of the design, and it also laid a foundation for building our own SoC system.

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